Research Statement

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Economic progress rests on the joint strength of markets and states—two systems that coordinate resources through very different mechanisms: prices and policies. Markets create prosperity when private incentives align with social welfare, and states sustain it when they can tax, regulate, and govern credibly. Yet in many economies—especially those marked by informality and institutional weakness—both falter in tandem. Markets are constrained by frictions that distort competition and investment, while states remain fiscally and politically fragile because much of economic activity lies beyond their reach. The result is a persistent equilibrium of low capacity and missed potential.

My research lies at this intersection. I ask a set of connected questions: *When can technology improve the state's ability to govern? When does it fail—and why? How do competitive forces within markets reinforce or undermine enforcement? And what role can citizens play in bridging the gap between state ambition and state capacity?* Guided by economic theory and tested through field and quasi-experimental methods, often in collaboration with governments and firms, I study how behavioral, technological, and organizational reforms can improve the functioning of markets and the capacity of the state.

Technology, Credibility, and the Limits of Digital State-Building

A foundational insight of my research is that information technologies—widely adopted as tools for "automated enforcement"—do not deliver durable compliance gains through deterrence unless the state can credibly act on the information they produce. In my job market paper with Isabelle Cohen (revise and resubmit, *Journal of Public Economics*), I study the rollout of a real-time digital reporting system that transmits point-of-sale transaction data directly to the tax authority. Using administrative VAT panels matched to device-level adoption records, we document three findings that challenge the optimistic narrative around digital tax infrastructure.

First, adoption is sharply selected: firms that integrate the technology are already more tax-engaged before they adopt, complicating naïve estimates of the technology's impact. Second, conditional on adoption, the immediate rise in reported sales and tax payments fades within months, as firms update their beliefs about the state's ability to act on the data; the increase that does persist is smaller and comes from the convenience of a stable digital workflow and the preferential rate rather than from deterrence. Revelation without follow-up teaches firms that monitoring alone is hollow: what lasts is a lower cost of compliance, not a greater fear of detection.

Third, and most novel, the reform's own incentive architecture undermined the tax instrument it was meant to strengthen. Preferential tax rates were linked to restrictions on input-credit claims, thereby shifting firms' reporting away from documenting input purchases. This weakened the self-enforcing chain of the VAT—the very mechanism that makes it, in theory, resistant to evasion. The lesson is not simply that "technology without teeth" fails; it is that poor design can actively erode the institutional logic of the tax system. Credibility and design—not technology alone—are central to digital state-building.

Informality Traps: When Markets Undermine the State

If the e-POS study shows that technology alone cannot sustain compliance, a natural question follows: what happens when the state adds enforcement? The answer, it turns out, depends on the market. In ongoing work with Michael Best, Anders Jensen, and Adnan Khan, I develop and test the concept of an *informality trap*—a dynamic equilibrium in which weak enforcement and market competition reinforce each other. When evasion is widespread, compliant firms face a cost disadvantage; they are undercut by competitors who do not remit taxes. This makes compliance individually irrational even when collective formalization would benefit everyone. The state, in turn, collects too little revenue to invest in credible enforcement, and the trap persists.

This project is among the first to test experimentally whether such traps exist. In a large-scale randomized enforcement intervention with tax authorities, we vary enforcement intensity across markets

to examine whether coordinated pressure can break the cycle. The design connects industrial organization—strategic complementarities, entry and exit—with core questions in public finance about the limits of state capacity: tax compliance is not merely an individual decision but a market-level outcome shaped by the competitive environment.

A companion project with the same team asks the question the trap makes urgent: who bears a tax that is never paid? A large share of the consumption-tax liability in developing economies is never remitted, and the evaded portion is a rent divided between the firms that conceal sales and the customers who take part. Which side captures it shapes first-order judgments in public finance: whether evading sellers in fact deliver lower prices—the premise beneath the view that consumption taxes are progressive in poor countries—and whose incomes an enforcement drive would touch. Existing evidence is indirect and market-level; this project measures the division directly. Punjab's restaurant tax supplies the laboratory: the statute taxes traceable payments at a reduced rate and cash at the full rate, so the wedge at stake on any bill is known exactly, and trained buyers with randomized payment instruments observe, purchase by purchase, what the customer saved, what the firm kept, and what the state received.

Automation and Bureaucratic Judgment

A parallel strand of my research asks how the state's own internal organization adapts to new technology. In ongoing work with Adnan Khan, Ben Olken, and Mahvish Shaukat, I conduct a randomized evaluation that integrates *computer vision* into property tax administration. Using satellite and street-level imagery, a machine-learning algorithm predicts property assessments and flags properties for reassessment. The experiment introduces these algorithmic benchmarks to a random subset of inspectors, allowing us to study how automation reshapes bureaucratic judgment, effort allocation, and accountability.

The question is more fundamental than whether AI improves accuracy: what happens to the human side of government when an algorithm can check—and challenge—a bureaucrat's work? Does it reduce discretion and improve consistency, or provoke strategic responses? Does it build citizen trust through transparency, or erode the relational accountability that existed before? These questions sit at the intersection of public economics, organizational economics, and the growing literature on AI in government, with direct implications for how developing countries automate administration.

Citizens as Monitors: Can Engagement from Below Strengthen Enforcement?

If monitoring without credibility is insufficient and enforcement from above depends on the market, a third project asks whether pressure from *below*—from citizens themselves—can complement formal enforcement. In an ongoing randomized evaluation with Isabelle Cohen, I provide the first experimental evidence on *consumer receipt lotteries*, a mechanism used globally (in Brazil, China, Taiwan, Portugal, and several U.S. states) but never rigorously evaluated.

The design assigns business eligibility to participate in a lottery that rewards consumers for submitting formal sales invoices. Leveraging administrative VAT data, firm and consumer surveys, and mystery-shopping audits, we examine whether small incentives motivate consumers to demand receipts, whether this reduces buyer-seller collusion, and whether the intervention generates spillovers to non-participating firms. The project directly tests whether citizen engagement can transform a monitoring technology—the same e-POS infrastructure I study in my job market paper—from a passive reporting tool into an active enforcement mechanism.

Social Identity and Inclusion in Markets

Markets do not operate in a social vacuum. Even as technology reduces transactional frictions, social norms and identity shape who participates, how they are treated, and what forms of exclusion persist. A key strand of my research examines these forces directly.

In a paper published in the *Journal of Economic Behavior and Organization*, I study racial discrimination in labor markets using a large-scale online experiment. I show that workers' productivity and willingness to cooperate depend not only on economic incentives but also on their employers'

social identity—revealing how social preferences can generate inefficiencies even in competitive settings. Extending this inquiry to digital markets, my paper in the *Journal of Development Economics* examines gender dynamics in an online marketplace. Using experimental buyer profiles that signal gender, I find that while women face no discrimination in prices or product quality, they experience substantially more unsolicited and flirtatious contact from male sellers—contact that, in conservative social contexts, imposes reputational and social costs on women.

These findings show that inclusion is not an automatic byproduct of market development or digitization; understanding who is excluded—and why—is essential for designing institutions that are both efficient and equitable.

Market Frictions, Climate Shocks, and Agricultural Networks

Just as institutional weaknesses limit how effectively states can govern, frictions and shocks within markets constrain economic performance. In ongoing work with Farah Said and Omar Gondal, I study how *information asymmetries* and *entrenched trading relationships* restrict firms' ability to respond to prices and realize potential gains from trade, leaving arbitrage opportunities unexploited.

A complementary project uses high-frequency trade data and remote-sensing imagery to examine how the catastrophic 2022 floods in Pakistan disrupted agricultural supply networks. We find sharp reductions in trade flows between rural producers and urban consumers; anticipatory effects—farmers increasing supply before the flood—appear only in historically high-risk regions, suggesting that prior experience shapes adaptation. The economic impact of climate events depends not only on physical damage but on the structure of the market institutions that mediate exchange.

Future Directions

Across these projects, my aim is a cumulative account of how markets and states co-evolve: gains from reducing frictions on the exchange side are durable only when paired with credible, low-cost enforcement, and stronger enforcement works only when it preserves competition and inclusion. The most ambitious extension of this agenda is a research-policy partnership, now being established with Pakistan's federal tax authority and collaborators in the United Kingdom, the United States, and Pakistan, to study the digitization pillar of the country's national tax transformation program. The authority is rolling out roughly two dozen reforms in a staggered sequence—mandatory digital invoicing, algorithmic risk-based audit selection, and high-powered incentives and peer ratings for tax officers, among others—creating a time-limited window to evaluate each causally with linked administrative data on taxpayers and officers. This partnership scales up the questions in my job market paper from a single provincial authority to a national administration and lets me study how monitoring technology, personnel incentives, and the enforcement process interact within a single system rather than in isolation.

Further projects deepen the citizen, inclusion, and market strands: one on the determinants of whistleblowing and how citizen reporting can be institutionalized as a credible accountability mechanism; one on customer-driven discrimination in digital marketplaces and the platform design features that mitigate it; and one with the Bank of Punjab, using the algorithmic eligibility rule of one of the world's largest state-subsidized revolving credit programs to ask whether the *form* of credit—a revolving line rather than a fixed-schedule term loan—changes how micro-enterprises manage working capital and grow.

More broadly, my ambition over the next decade is to move from documenting when reforms succeed or fail toward a predictive framework for *institutional design*—one that specifies the conditions under which combinations of technology, enforcement, and citizen engagement deliver durable improvements in state capacity and market performance. The evidence base I am building—field experiments, administrative panels, and frontier measurement tools—is designed to support exactly that. *(Updated August 2026)*